# *MoMa-Kitchen*: A 100K+ Benchmark for Affordance-Grounded Last-Mile Navigation in Mobile Manipulation

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## Abstract

*Existing navigation benchmarks for embodied agents primarily reward reaching or looking at a target, but they do not directly evaluate whether the robot's final base pose makes manipulation feasible. This gap is especially important in mobile manipulation, where a small pose error can turn a reachable target into a failed grasp because of collisions, articulated-object kinematics, or embodiment-specific workspace limits. We introduce MoMa-Kitchen, a benchmark for affordance-grounded last-mile navigation in kitchen environments. MoMa-Kitchen contains 127,343 episodes over 569 procedurally generated scenes, with egocentric RGB-D observations, aligned scene and floor point clouds, robot metadata, and dense floor affordance maps derived from automated manipulation rollouts. The task is to predict a continuous robot-conditioned affordance map over feasible base locations rather than a single waypoint near the target. We also define an evaluation protocol that combines dense map fidelity metrics with Manipulation Success Rate (MSR) to test downstream usefulness. On unseen scenes, the released reference baseline NavAff achieves the best performance among the compared point-cloud baselines, reaching 0.147 RMSE, 0.0115 logMSE, 0.680 PCC, and 0.696 similarity, with Top1 and Top5 MSR of 0.72 and 0.66, respectively. The benchmark remains far from saturated, exposes large differences across architectural choices, and shows that global scene geometry matters more than a coarse robot descriptor in the tested setting. Qualitative real-world experiments further suggest that simulation-trained affordance predictors can produce plausible base-placement maps from commodity RGB-D sensing. These results position MoMa-Kitchen as a diagnostic benchmark for measuring the manipulation relevance of the final navigation decision, rather than only the ability to approach a target.*


## 1. Introduction

Embodied agents are increasingly evaluated on navigation, language grounding, and rearrangement tasks in realistic 3D environments [1, 2, 5, 9, 17]. Yet a persistent failure mode in mobile manipulation appears in the final meters before interaction: a robot may successfully approach the target object, but still stop at a base pose that makes the actual manipulation attempt infeasible. In kitchens, for example, the correct pose depends not only on target proximity but also on reachability, obstacle clearance, articulation direction, and robot morphology.

Figure 1. Conventional navigation objectives often stop once the robot is close to the target, while mobile manipulation depends on whether the final base pose affords interaction. MoMa-Kitchen makes this last-mile decision the object of evaluation.

This "last-mile" problem is weakly captured by standard navigation metrics and only indirectly measured by end-to-end success. ObjectGoal, vision-and-language navigation, and scene-level embodied benchmarks emphasize exploration, localization, or semantic grounding [1, 3, 5, 9, 10, 14]. Mobile manipulation and rearrangement benchmarks evaluate integrated systems [2, 13, 17], but they typically do not provide dense supervision over the space of feasible base poses. As a result, it is hard to isolate whether failure is caused by perception, long-range navigation, or the final approach geometry.

We present MoMa-Kitchen, a benchmark for affordance-grounded last-mile navigation in mobile manipulation. Instead of predicting a single goal location, the benchmark asks a model to produce a dense floor affordance map conditioned on scene geometry and robot embodiment. Ground-truth affordance values are obtained automatically by executing manipulation attempts from sampled base poses in simulation and then densifying these sparse success labels over the floor geometry. This formulation converts the final navigation decision into a measurable supervised prediction

problem while preserving its downstream relevance through a manipulation-based evaluation protocol.

The benchmark is built in a procedural kitchen simulator using BestMan [16] with articulated assets from PartNet-style resources [6, 15]. It contains 127,343 episodes across 569 scenes, with train/test splits over disjoint scenes. The benchmark description reports 137 asset types, including rigid targets, articulated targets, and obstacles, and varies robot morphology to induce different reach and clearance constraints. Each episode provides first-person RGB-D, aligned point clouds, and a dense floor affordance map for a target object.

Beyond the dataset itself, we provide a benchmark protocol and a released reference model, NavAff, that operates on processed geometric observations. The key empirical findings are consistent across the provided tables and figures: NavAff outperforms adapted 3D point-cloud baselines on dense affordance prediction and downstream manipulation success, but performance remains well below saturation; global scene context and attention-based fusion are more important than the coarse robot descriptor used here; and real-world qualitative experiments indicate plausible transfer despite the benchmark being simulation-generated.

The central design choice in MoMa-Kitchen is to treat feasible base placement as a dense spatial prediction problem rather than as a single successful trajectory endpoint. This makes the benchmark more diagnostic. A model can be penalized for assigning high score to nearby but obstructed floor regions, rewarded for identifying narrow feasible corridors, and analyzed slice-by-slice without rerunning a full navigation stack. The benchmark is therefore intended as a complementary evaluation layer for mobile manipulation systems rather than a replacement for end-to-end household benchmarks.

Our contributions are threefold. First, we introduce a large-scale benchmark for last-mile navigation grounded in manipulation feasibility rather than mere target proximity. Second, we provide a fully automated data-generation and annotation pipeline that yields dense floor affordance supervision without human labeling. Third, we establish an evaluation protocol and baseline results that make the benchmark diagnostic, reproducible, and useful for future comparison.

## 2. Related Work

**Navigation benchmarks.** Large-scale embodied navigation benchmarks have driven progress in scene understanding and goal-reaching, including Room-to-Room [1], REVERIE [9], Gibson [14], Matterport3D [3], HM3D [10], and HSSD-200 for ObjectGoal navigation [5]. These benchmarks are valuable for measuring exploration and generalization, but they do not directly score whether the final pose affords manipulation. MoMa-Kitchen complements them by focusing on the geometric and embodiment-specific decision that remains after the object has already been localized.

**Mobile manipulation and rearrangement benchmarks.** Rearrangement and open-vocabulary mobile manipulation benchmarks evaluate integrated perception-planning-control systems [2, 13, 17]. They are closer to household robotics than pure navigation tasks, but supervision is typically sparse and end-to-end. In contrast, MoMa-Kitchen isolates a narrower but important subproblem: selecting a manipulation-feasible base pose in cluttered kitchens. Recent task-aware placement methods such as MoMa-Pos [11] and pose-aware navigation methods such as Navi2Gaze [19] also highlight the importance of robot pose, but they are method papers rather than benchmarks with dense annotations and standardized evaluation.

**Affordance-grounded representations.** Affordances and value maps have been used to guide robot interaction and planning [4, 12]. These works motivate representing action feasibility spatially rather than as a discrete class label. MoMa-Kitchen translates that idea into a benchmark: it provides dense, robot-conditioned floor affordance targets backed by simulation rollouts, making it possible to compare models on a common evaluation protocol without conflating the benchmark contribution with a particular planning algorithm.

## 3. Benchmark Design

### 3.1. Design Principles

MoMa-Kitchen is designed around four principles. First, the labels should be *manipulation-grounded*: high affordance means that a manipulation attempt succeeds from that pose, not merely that the target is visible or nearby. Second, the benchmark should be *embodiment-sensitive*: the same scene can induce different feasible floor regions for robots with different reach and height constraints. Third, it should be *dense and diagnostic*: every episode should expose a continuous feasibility landscape rather than a single binary success outcome. Fourth, it should be *reproducible*: the entire pipeline, from scene generation to label construction, should be automated and deterministic.

### 3.2. Task Definition

Each episode in MoMa-Kitchen is defined by a kitchen scene, a target object, a robot embodiment, and an egocentric observation. The benchmark provides first-person RGB-D, a global point cloud, a floor point cloud, and robot metadata. The prediction target is a dense affordance value over floor locations near the target. Intuitively, high values indicate that stopping at that base pose is likely to enable

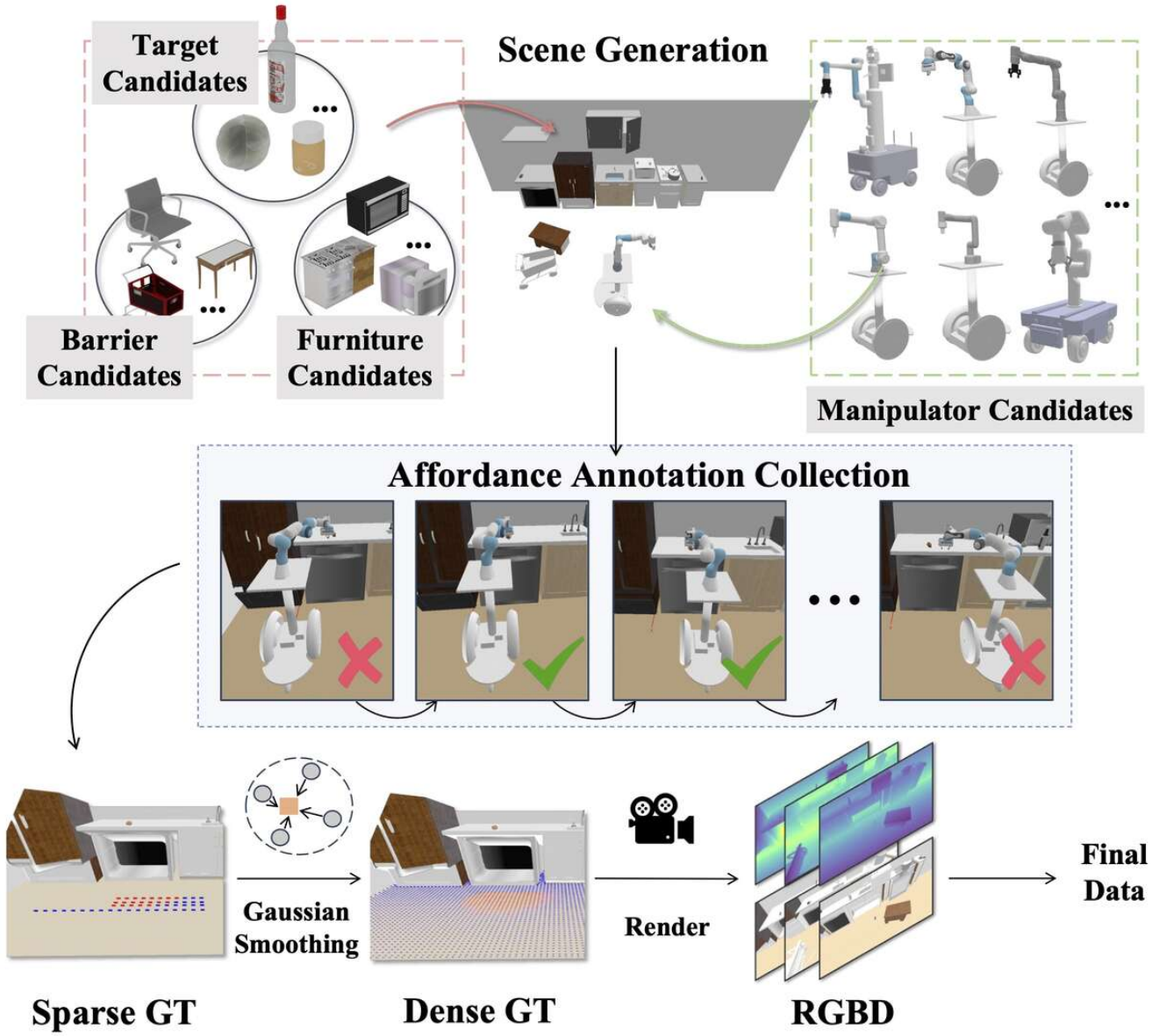


Figure 2. Overview of MoMa-Kitchen. The benchmark procedurally constructs kitchen scenes, samples robot viewpoints and candidate base poses, executes manipulation attempts to obtain sparse success labels, and densifies those labels into continuous floor affordance maps used for training and evaluation.

successful interaction with the target under the robot's reach and collision constraints.

This formulation differs from point-goal navigation and target gazing. The goal is not simply to reach the nearest point to the object, nor merely to orient the camera toward it. Instead, the task is to estimate the robot-conditioned feasibility landscape of the floor around the object. We treat this as a benchmark and evaluation protocol for last-mile base placement, not as a full-stack mobile manipulation benchmark.

### 3.3. Scene Construction and Data Collection

MoMa-Kitchen is generated in BestMan, a PyBullet-based mobile manipulation platform [16]. The benchmark uses procedurally varied kitchen layouts populated with rigid and articulated assets from PartNet-style resources [6, 15]. The benchmark description reports 569 scenes and 127,343 episodes in total, covering 137 asset types split across rigid targets, articulated targets, and obstacles.

For each scene configuration, the pipeline samples targets, arranges obstacles to induce clutter, selects a robot

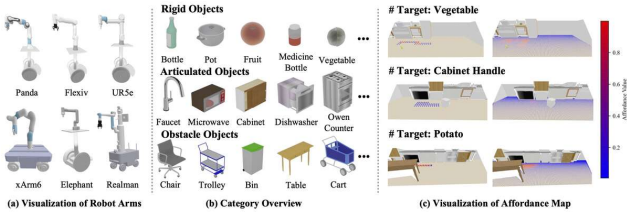


Figure 3. Examples of benchmark assets and annotations. The provided visualization highlights robot embodiments, first-person observations, and floor-level affordance supervision aligned with the scene geometry.

Table 1. **Dataset Split Statistics.**

| Split | Scenes | Configurations | Episodes |
|---|---|---|---|
| Train | 456 | 11,408 | 102,687 |
| Test (Unseen Scenes) | 113 | 2,747 | 24,656 |

embodiment, and mounts an egocentric RGB-D camera on the robot arm. Viewpoints are then sampled around the target until the target is visible, and the system records RGB, depth, and aligned 3D geometry. Candidate base poses are sampled over a semicircular region on the floor whose radius is determined by the arm's reachable workspace. At each sampled pose, the simulator attempts the relevant manipulation and records binary success or failure. These sparse labels are then transformed into the robot base frame, associated with nearby floor points, and densified into a continuous floor affordance map using Gaussian interpolation over nearest neighbors. Viewpoints without line of sight are rejected, and the manipulation trials are obstacle-aware, so the labels encode both reachability and collision constraints.

The resulting labels are richer than a grasp/no-grasp annotation for a single viewpoint. They express a continuous function over the floor, which allows researchers to study ranking quality, calibration, and the geometry of failure. This matters in cluttered kitchens where feasible poses often form thin regions in front of articulated appliances or along obstacle boundaries. Figure 3 illustrates the benchmark's emphasis on robot embodiment and aligned floor-level supervision.

### 3.4. Benchmark Splits and Inputs

Table 1 summarizes the train/test split. The test set contains 113 unseen scenes, so evaluation measures generalization beyond memorized scene layouts. No human annotators are involved: all labels are produced automatically by a deterministic simulation pipeline.

In addition to the split statistics, the released data organization exposes the supervision interface used by the reference implementation: transformed scene point clouds, floor point clouds, point subsets corresponding to the target and nearby barriers, and Gaussian-smoothed floor affordance labels. This makes it possible to benchmark methods that operate on raw RGB-D, processed geometry, or both, while keeping the evaluation target fixed.

## 4. Benchmark Validation

**Physical grounding of labels.** The benchmark labels are not heuristic distances to the target. They originate from executed manipulation attempts at sampled base poses, then are smoothed into a dense field. This matters because the feasible region can be sharply shaped by articulated motion, occlusion, and clutter, all of which are poorly approximated by Euclidean proximity alone. The floor maps therefore encode a stronger notion of task relevance than typical navigation targets.

**Reproducibility and annotation quality.** Because annotation is fully automated, there is no inter-annotator agreement to report. The tradeoff is different: fidelity depends on simulator realism, camera calibration, and geometric alignment. MoMa-Kitchen addresses this through a deterministic pipeline with fixed intrinsics, explicit coordinate transforms, target visibility checks, and collision-aware rollout labels. The public code also reveals the concrete supervision artifacts used by the released baseline, including floor point clouds, transformed scene point clouds, target and obstacle point subsets, and Gaussian-smoothed affordance labels.

**Difficulty and diagnostic value.** The benchmark is neither trivial nor saturated. Random candidate selection yields very poor downstream manipulation success, while learned methods improve substantially yet still leave large gaps between dense prediction metrics and perfect success. The benchmark also discriminates between architectural choices: in the ablations, removing global scene context or the fusion module causes notable performance drops, showing that the task is sensitive to the geometry needed for last-mile reasoning rather than being solved by local heuristics alone.

**Validity of the evaluation protocol.** The protocol is useful only if dense-map quality relates to actual task outcomes. The benchmark supports this connection in two ways. First, the labels themselves are derived from manipulation rollouts rather than proxy geometry. Second, the downstream MSR results preserve the method ranking seen in dense metrics: the best map predictor, NavAff, is also the best performer on Top1 and Top5 MSR. This agreement is not a proof of perfect metric validity, but it is evidence that

the benchmark measures a behaviorally meaningful quantity.

# 5. Experimental Protocol

## 5.1. Metrics

We evaluate predicted affordance maps on the test split using RMSE, logMSE, Pearson correlation coefficient (PCC), and similarity (SIM). In the released evaluation, SIM corresponds to cosine similarity between flattened predicted and ground-truth maps. These metrics assess dense prediction fidelity over valid floor points.

To connect dense prediction quality to actual task utility, we additionally report Manipulation Success Rate (MSR). MSR measures success after selecting high-scoring base poses from the predicted affordance map and attempting manipulation from those locations. We report Top1 and Top5 settings corresponding to the highest-ranked candidate and the five highest-ranked candidates, respectively. Because Top5 reflects ranking quality across multiple candidates rather than an "any-success" oracle, it can be lower than Top1.

## 5.2. Baselines

We compare the released reference baseline NavAff against adapted point-cloud baselines: PointNet++ [7], VoteNet [8], and H3DNet [18]. We also report a random policy for MSR. These baselines are not originally designed for dense floor-affordance regression, so the comparison should be interpreted as a benchmark reference rather than a definitive model ranking across all possible architectures.

## 5.3. Released Reference Model

The released NavAff implementation is a lightweight geometric baseline. It encodes a global point cloud and a floor point cloud using PointNet++-style hierarchical set abstraction [7], augments global scene points with target/obstacle indicators, and injects a compact robot descriptor through an MLP. An attention-based fusion block combines the coarse floor representation with global context and robot information, after which a point-wise decoder predicts affordance values on floor points. Training uses a weighted mean squared error objective over valid floor points. In the released configuration, training uses batch size 64 for 6 epochs.

## 5.4. Implementation Details

The public training code uses a single unseen-scene test split, a batch size of 64, Adam optimization, cosine learning-rate decay, and a weighted MSE objective that downweights a subset of zero-affordance floor points. The released data loader pads floor and global point clouds to fixed sizes and attaches a one-dimensional semantic feature

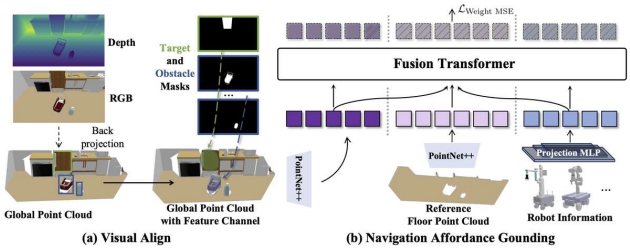


Figure 4. Released reference baseline, NavAff. The model combines floor geometry, global scene context, and a compact robot descriptor to predict dense floor affordances. In this paper it serves as a benchmark reference, not as the main contribution.

to global points to mark target and obstacle regions. These details matter because they define the reference evaluation setting future work should reproduce before claiming gains on the benchmark.

# 6. Results and Analysis

## 6.1. Main Results

Table 2 reports dense affordance prediction performance. NavAff is best on all four metrics, improving over the strongest baseline, PointNet++, from 0.164 to 0.147 RMSE and from 0.565 to 0.680 PCC. The gap is larger over H3DNet and VoteNet. This suggests that explicit conditioning on floor geometry and task-relevant context is more suitable for this benchmark than repurposing generic 3D detection pipelines.

Table 2. **Main results.** Quantitative evaluation of navigation affordance grounding performance of NavAff.

| Method | RMSE ↓ | logMSE ↓ | PCC ↑ | SIM ↑ |
|---|---|---|---|---|
| PointNet++ [7] | 0.164 | 0.0142 | 0.565 | 0.589 |
| VoteNet [8] | 0.167 | 0.0143 | 0.543 | 0.570 |
| H3DNet [18] | 0.174 | 0.0156 | 0.503 | 0.522 |
| NavAff | **0.147** | **0.0115** | **0.680** | **0.696** |

Table 3 shows downstream MSR. NavAff again performs best, reaching 0.72 Top1 and 0.66 Top5. The random baseline is very poor, confirming that the success landscape is far from uniform over candidate poses. A notable pattern is that Top5 is lower than Top1 for every method. This indicates that the benchmark is sensitive to the ordering of the highest-scoring candidates: finding one excellent pose is easier than maintaining precision over the top of the ranking.

Taken together, the two tables show that MoMa-Kitchen does more than reward smooth regression. Methods that better reconstruct the dense affordance landscape are also the ones that choose better final poses for manipulation. At the same time, the remaining gap to perfect MSR indicates

Table 3. **Manipulation Success Rate (MSR).**

| MSR | Random | H3DNet | VoteNet | PointNet++ | NavAff |
|---|---|---|---|---|---|
| Top1 | 0.080 | 0.54 | 0.56 | 0.60 | **0.72** |
| Top5 | 0.046 | 0.47 | 0.53 | 0.58 | **0.66** |

Table 4. **Ablation study results of NavAff.**

| | RMSE ↓ | logMSE ↓ | PCC ↑ | SIM ↑ | top1MSR ↑ |
|---|---|---|---|---|---|
| NavAff | 0.147 | 0.0115 | 0.680 | 0.696 | 0.72 |
| *w/o* robot information | 0.148 | 0.0115 | 0.670 | 0.688 | 0.70 |
| *w/o* VAM | 0.165 | 0.0140 | 0.562 | 0.589 | 0.63 |
| *w/o* global point cloud | 0.168 | 0.0144 | 0.534 | 0.568 | 0.58 |

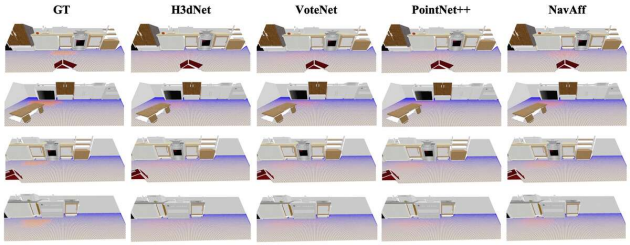


Figure 5. Qualitative comparison of predicted floor affordances. NavAff more closely matches the shape and concentration of high-affordance regions in the provided examples, while competing baselines are noisier or overly diffuse.

that accurate ranking of a few crucial high-affordance locations remains difficult, especially in narrow feasible regions.

## 6.2. Ablations

Table 4 evaluates the released baseline components. Removing the attention-based fusion module labeled "VAM" in the table reduces PCC from 0.680 to 0.562 and Top1 MSR from 0.72 to 0.63. Removing the global point cloud is even more damaging, dropping PCC to 0.534 and Top1 MSR to 0.58. By contrast, removing the robot descriptor causes a much smaller change, from 0.680 to 0.670 PCC and from 0.72 to 0.70 Top1 MSR. In the current setup, scene geometry therefore appears to dominate the benchmark signal, while the coarse robot metadata used by the released baseline has a secondary effect.

This ablation pattern is itself a useful benchmark finding. It suggests that, at least for the current representation, most of the difficulty lies in interpreting global layout constraints rather than in exploiting the low-dimensional robot metadata. That does not mean embodiment is unimportant in principle. Rather, it indicates that the present robot encoding is probably too coarse to unlock the full benchmark variation, leaving room for future methods that model embodiment more explicitly.

## 6.3. Qualitative Analysis and Real-World Transfer

Figure 5 illustrates why dense-map metrics matter. The main errors are not only score calibration errors, but also geometric ones: several baselines spread probability mass to side or obstructed regions that are close to the target but not manipulation-feasible. This is precisely the failure mode the benchmark is designed to expose.

We consistently observe three qualitative error modes in the provided examples. First, some methods over-score points that are visible from the camera but unreachable because of object articulation or side approach geometry. Second, some produce overly diffuse predictions, which can still yield reasonable RMSE but degrade ranking quality and Top5 MSR. Third, some fail to account for nearby barriers and assign high score to poses that are close to the target but collision-prone. These modes support the claim that MoMa-Kitchen probes more than target localization.

**How to interpret scores.** The qualitative examples also clarify why the benchmark reports both dense-map metrics and MSR. Dense metrics measure how well a model reconstructs the overall feasibility landscape, but deployment depends disproportionately on whether the very highest-ranked poses are executable. Two methods can therefore have similar RMSE while inducing different manipulation outcomes if one is better calibrated around a small number of sharp affordance peaks. For benchmark users, this distinction is useful: it separates errors in global map estimation from errors in top-of-ranking decision quality.

The provided real-world experiment further tests whether simulation-trained affordance predictors can still produce sensible maps from real RGB-D sensing. As shown in Figure 6, the sim-trained model produces plausible high-affordance regions in front of reachable objects when coupled with commodity depth sensing and open-vocabulary segmentation. We treat this as qualitative evidence only; no real-world quantitative benchmark numbers are reported here.

## 7. Discussion

MoMa-Kitchen measures one specific but important capability: predicting whether a final base pose will support manipulation. It does not measure long-horizon exploration, language understanding, full grasp synthesis, or closed-loop recovery after failed execution. Strong performance on MoMa-Kitchen should therefore be interpreted as evidence of better last-mile base-placement reasoning, not as a complete measure of mobile manipulation competence.

The benchmark also has limitations. First, the data is synthetic and kitchen-centric, so ecological validity depends on the realism of the simulator, assets, and physics. Second, the dense affordance maps are interpolated from

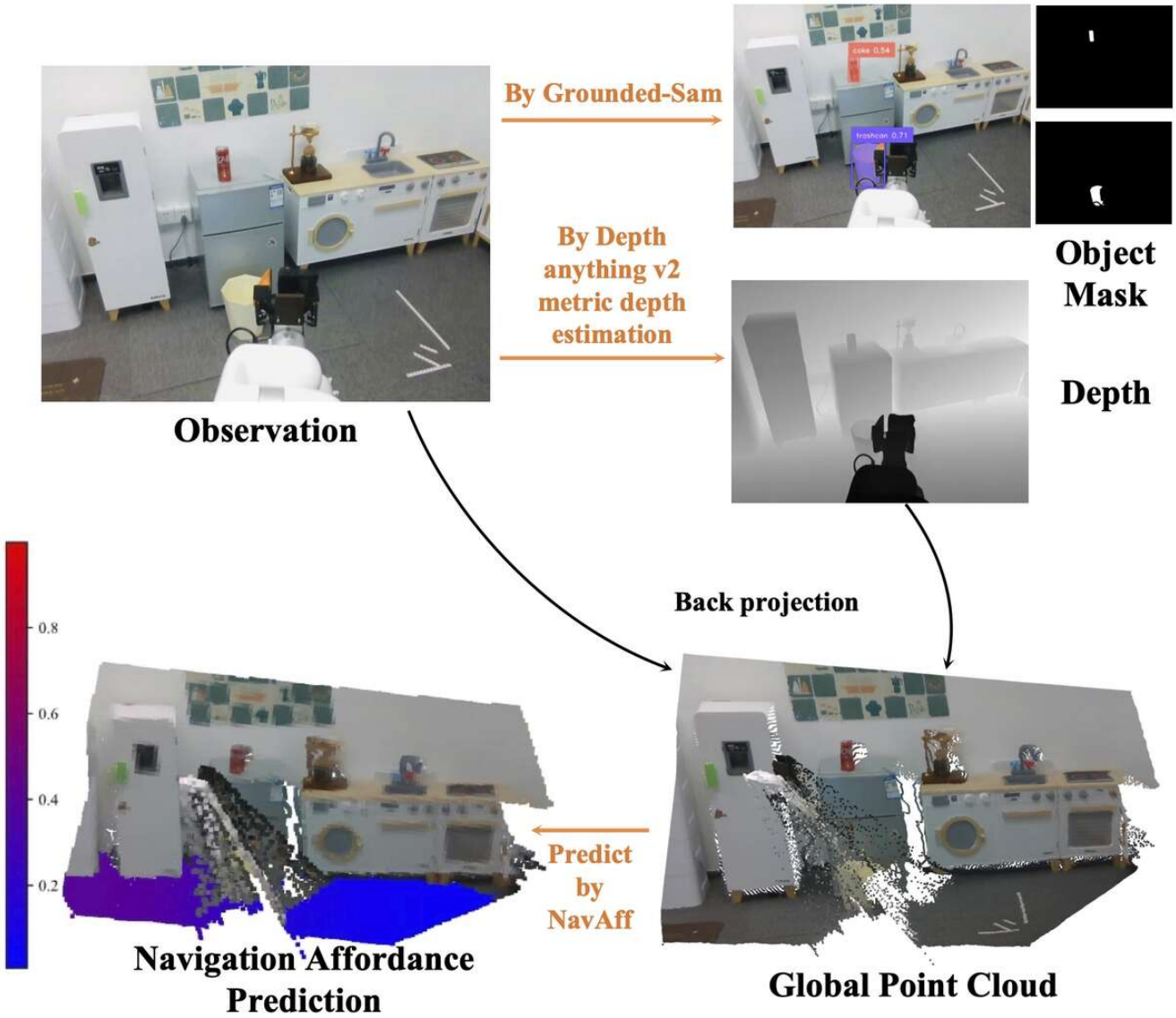


Figure 6. Qualitative real-world application pipeline. A model trained on MoMa-Kitchen can be paired with RGB-D sensing and open-vocabulary segmentation to generate plausible base-placement affordances in real kitchens, although the current evidence is qualitative rather than a standardized real-world benchmark.

sparse rollout labels, which improves supervision density but can smooth sharp feasibility boundaries. Third, the released reference model operates on processed geometric inputs rather than raw RGB-D end to end, leaving richer multimodal baselines for future work. Finally, the real-world evidence is qualitative, so sim-to-real generalization remains an open evaluation problem rather than a solved one.

These limitations do not reduce the usefulness of the benchmark; they clarify its intended role. MoMa-Kitchen is most valuable as a controlled, reproducible evaluation suite for studying the final navigation decision that precedes manipulation.

## 8. Conclusion

We introduced MoMa-Kitchen, a benchmark for affordance-grounded last-mile navigation in mobile manipulation. By supervising dense robot-conditioned floor affordance maps and evaluating them with both map-fidelity metrics and manipulation success, the benchmark measures a failure mode that is largely invisible to standard navigation benchmarks. The released results show that the task is challenging, discriminative, and not saturated, while the ablations indicate that global scene geometry is

especially important for success. We hope MoMa-Kitchen helps make evaluation of mobile manipulation systems more diagnostic by explicitly testing whether the robot stops where manipulation is actually possible.

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